

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Python** agentic workflows using **LangGraph** for a mission-critical financial operations dashboard, processing ~2,800 transaction messages per hour.
- Developed and operated **RAG pipelines** to ingest and retrieve data from structured, semi-structured, and unstructured sources, supporting complex analytical queries.
- Engineered context management strategies for multi-step and long-running agent executions, ensuring stateful orchestration and reducing processing errors by **18%**.
- Built and maintained **Python** scripts for automating ETL processes and scheduled reporting on enterprise business systems, improving data refresh cycles by **30%**.
- Contributed to a **Java Spring Boot** microservice, developing REST APIs for cross-system data exchange and integrating with enterprise ERP modules.
- Managed production incidents for enterprise applications using **Splunk** and **Dynatrace**, resolving P1/P2 issues within defined SLA targets.

SKILLS

AI / ML – AI, Agentic, RAG, LangGraph, Python, scikit-learn, pandas, NumPy

Backend Development – Python, Java, Spring Boot, REST APIs, Flask, FastAPI

Data Management – SQL, PostgreSQL, MySQL, Oracle, Data Warehousing, ETL, Power Query

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Jenkins, Git, Linux

BI & Analytics – Power BI, Tableau, DAX, Data Modelling, R / RStudio, Statistical Analysis

PROJECTS

AI-Powered Transaction Monitoring System (HCL Technologies)

- Developed a **Python**-based agentic system for real-time transaction monitoring, integrating **LangGraph** for workflow orchestration and state management.
- Implemented **RAG** architecture to query historical transaction data and identify anomalous patterns, improving detection accuracy by **15%** over previous methods.
- Designed context management for long-running anomaly detection workflows, ensuring data consistency and reducing manual triage effort for **~400 alerts** weekly.

Automated ERP Data Integration Pipeline (HCL Technologies)

- Created **Python** scripts to automate data extraction, transformation, and loading from legacy ERP systems into a central PostgreSQL database.
- Developed data validation checks and error handling mechanisms, reducing data integrity issues in downstream reports by **22%**.
- Built scheduled jobs using **Linux cron** and integrated with **Jenkins** for CI/CD, ensuring reliable daily data updates for business analysts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)